Lecture 5: Big Data: Focus on Cancer Genomic Data

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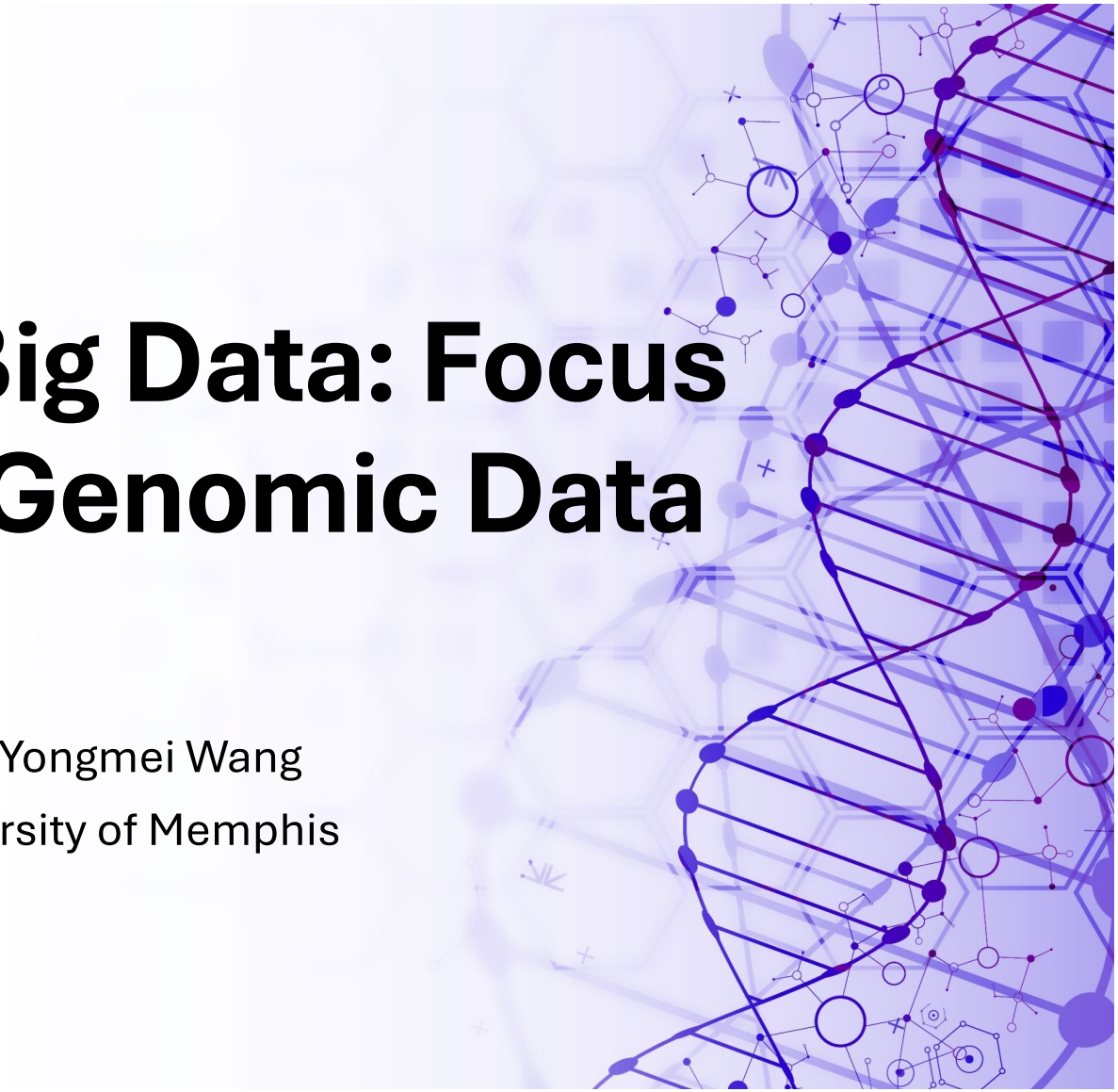
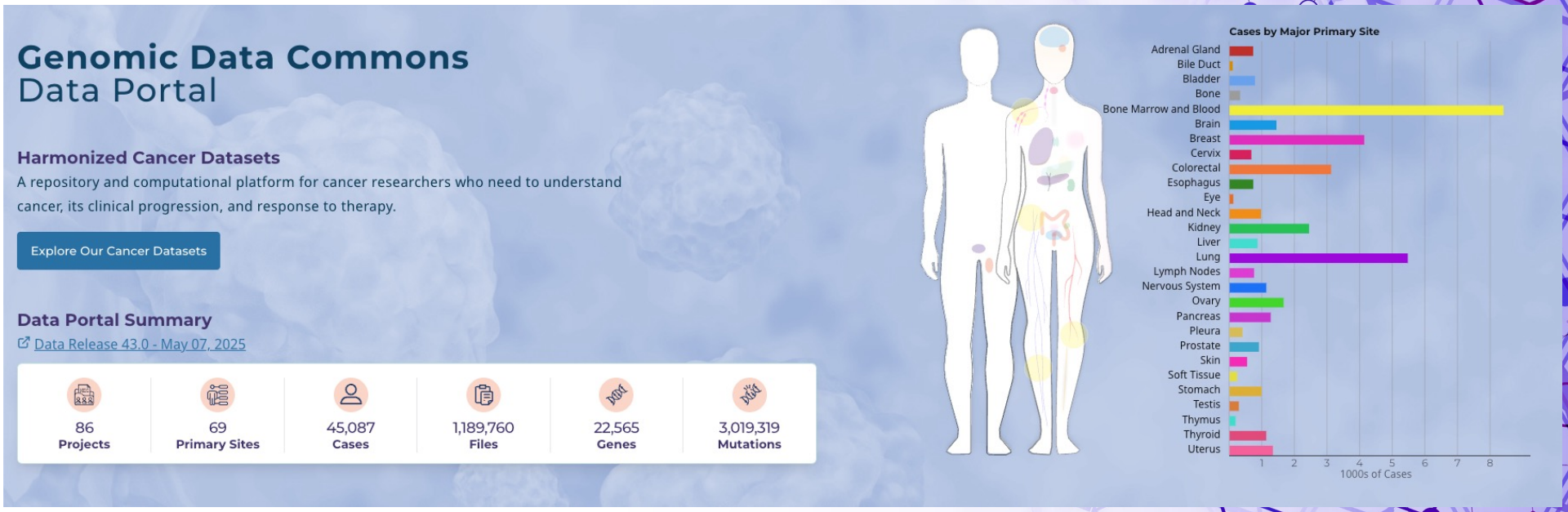
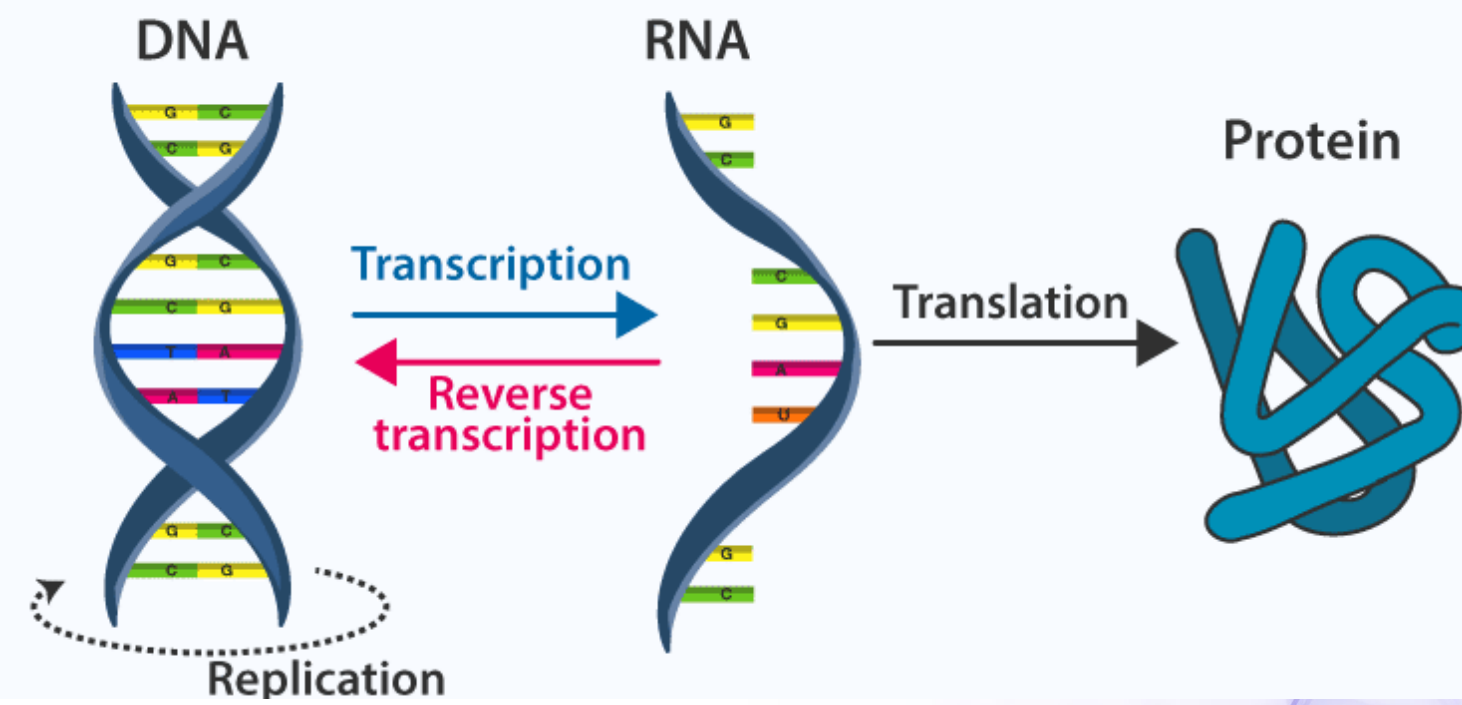
The University of Memphis



What have we done so far?



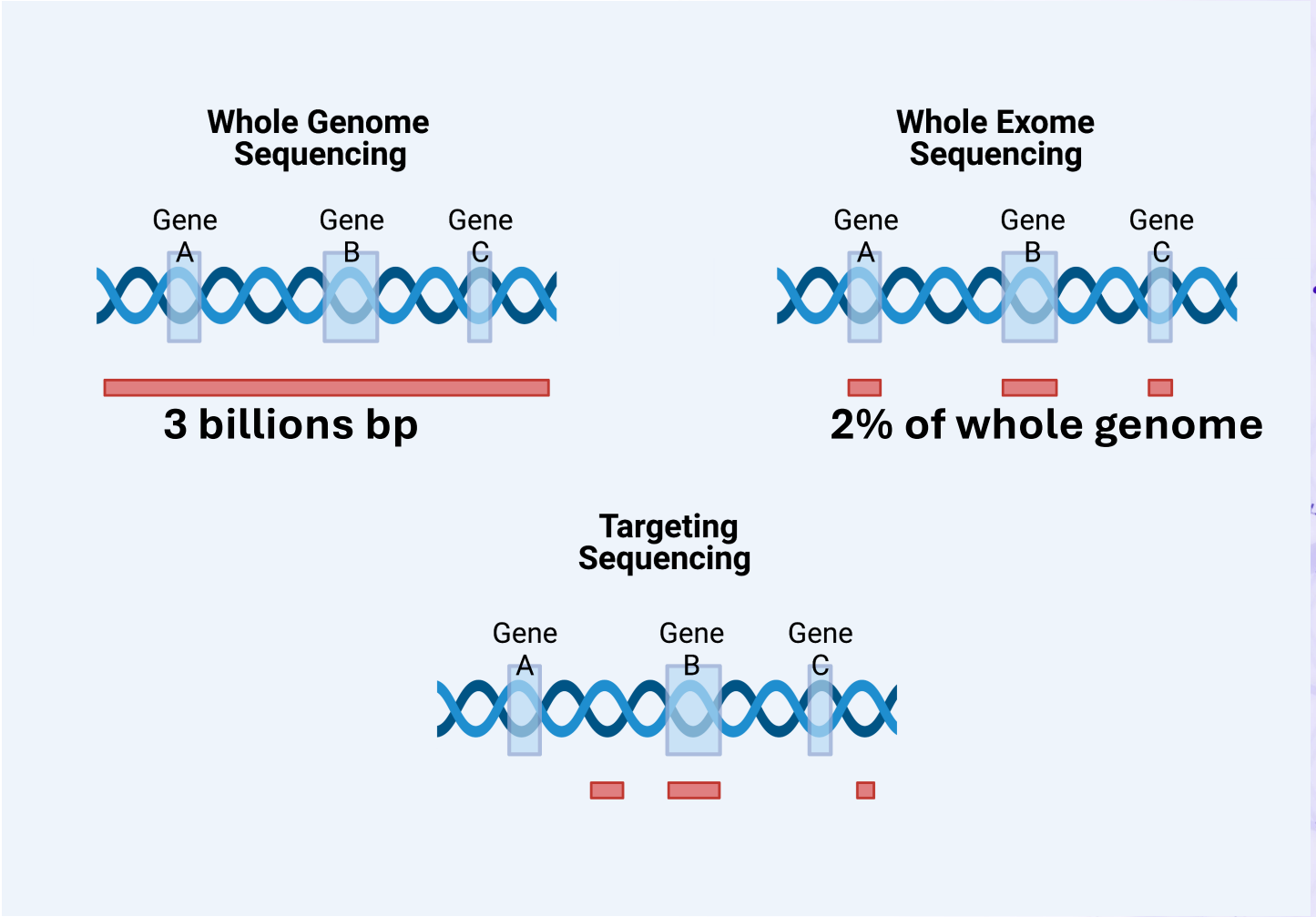
Genomic Data Commons (GDC) https://portal.gdc.cancer.gov/
Screen captured on July 2025.



Genomic Data

Transcriptomic Data --
Related to gene expression

Proteomic Data



https://www.fjc.gov/content/361254/dna-sequencing-and-what-it-can-reveal-about-dna-variation

Whole genome sequencing –workflow

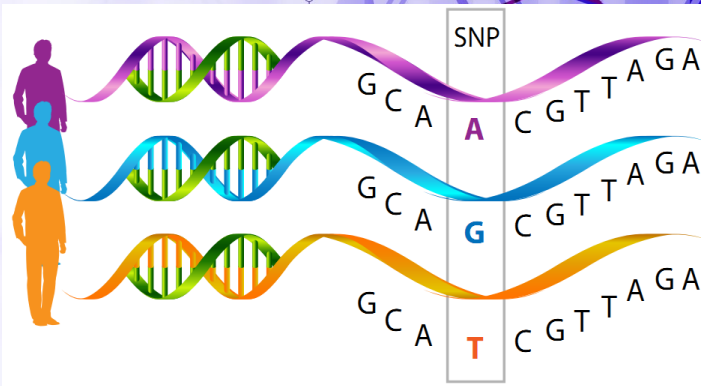
- First watch a high-level overview of whole genome sequencing: from samples to report (https://www.youtube.com/watch?v=rH6vsoyrc_U)
  - Samples (tumors, blood samples)
  - DNA purified /processed
  - Sequence reads fed into bioinformatics pipelines for alignments
  - Variation Annotation Calling

What you get from genomic data?

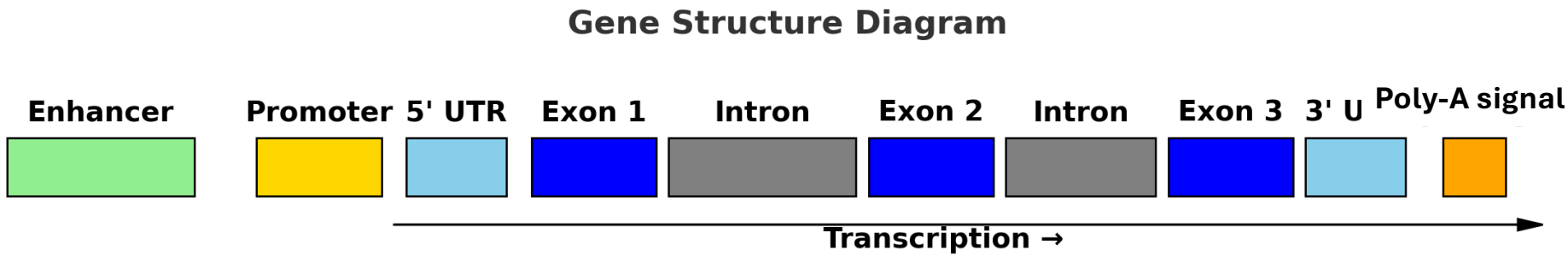
- After alignment of the sample's genome against the reference human genome, you will then do Variant Calls
- There are several types of "Variants".
  - SNPs (Single Nucleotide Polymorphism): **SNP** is a variation at a **single position** in the DNA sequence among individuals.
  - INDELs (insert or deletion)
  - CNV (copy number variations)
  - Structural Variants
- Variants are further divided
  - **Germline variants** –detected by sequencing normal samples DNA (blood or saliva)
  - **Somatic variants** are found by comparing **tumor DNA** to matched normal DNA from the same patient: If variants are found in tumor, but not in normal, it is **somatic variants**

What is SNPs

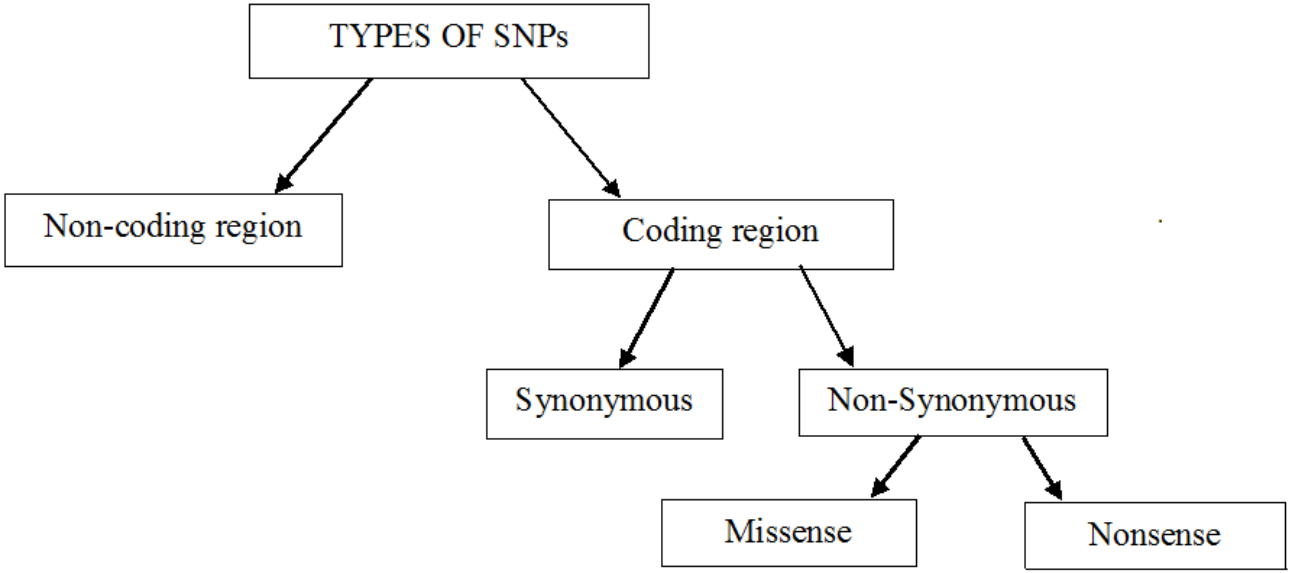
- SNPs are single-nucleotide polymorphisms: variations in one nucleotide.
- The human genome is known to have over 500 million SNPs
- Estimated common SNPs in humans: ~10–15 million. These are used for population genetic study.
- Each individual may carry 4~5 million SNPs (human to human genetic variation is ~0.1% of the entire genome);
- A lot of SNPs locate in non protein-coding region, but some fall on protein-coding regions; this can lead to a change of amino acids



Gene in genome



- A gene on genome is **not limited to the sequence that codes for a protein**.
- It includes both **coding and non-coding regions**, such as **exons** and **introns**.
- **Exons** are transcribed and retained in the mature mRNA — and typically translated into protein.
- **Introns** are transcribed but removed during **RNA splicing**.
- **Through alternative splicing**, different combinations of exons can be joined together, resulting in **multiple mRNA and protein variants** from a single gene.



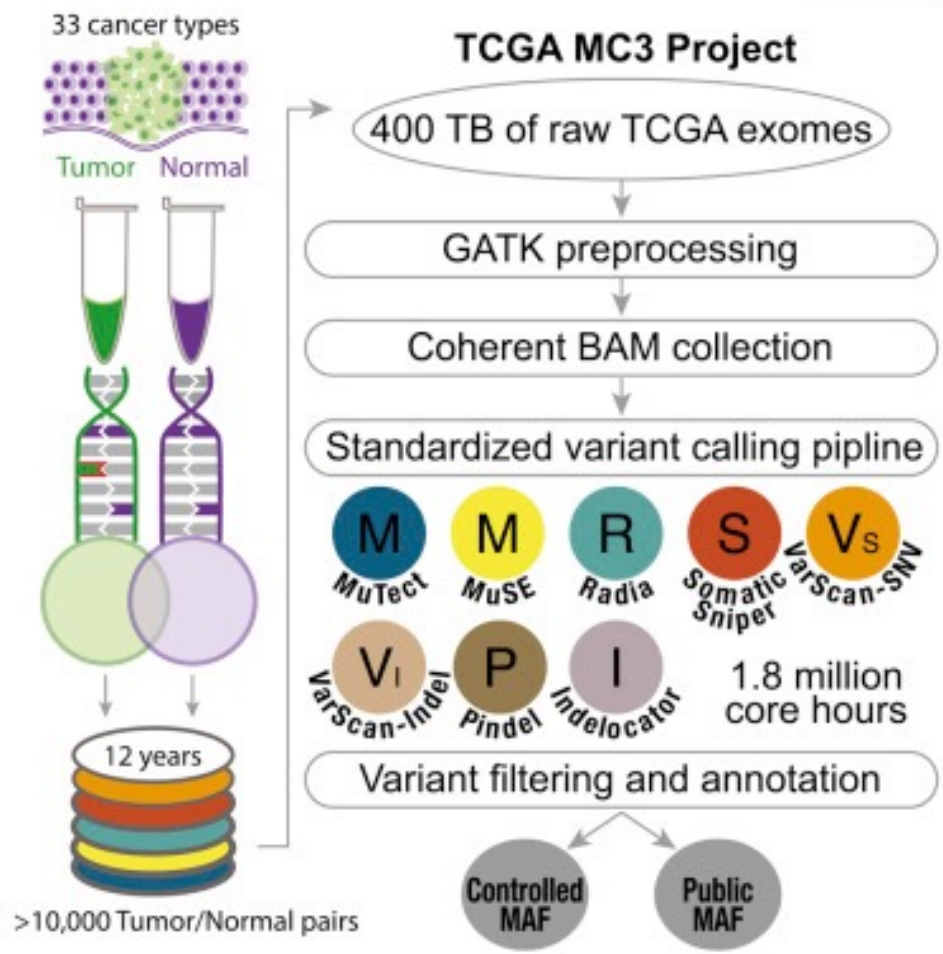
Synonymous: means the amino acid did not change

Non-Synonymous: alters the amino acid. Missense: change to a different amino acid; Nonsense: change to a STOP codon

Somatic variants

- Somatic variants are called when compare “tumor samples genome” against patient-matched “normal samples”
- In contrast, Germline SNPs refer to variants present in patient normal samples, not tumor samples. It could be patient’s blood or normal tissues around tumors
- Determining SNPs in tumor samples takes a lot of computational effort
- Check the reference: https://www.cell.com/fulltext/S2405-4712(18)30096-6

Bioinformatics Pipelines



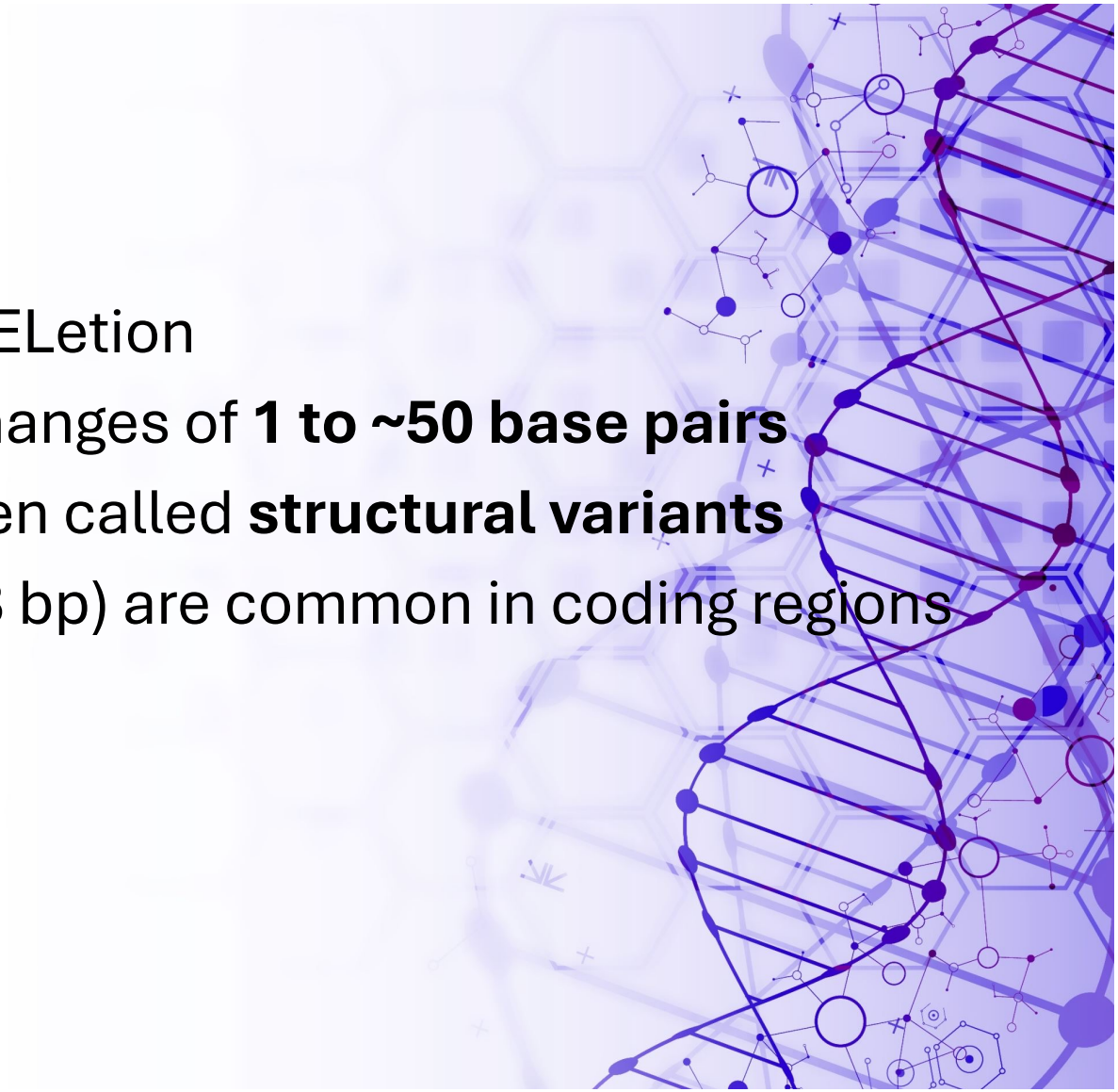
Datasets are accessible at Xena Browser

For example:
TCGA BRCA: under somatic mutation (SNPs and INDEL): MC3 gene-level files

These are processed Mutation Annotated Files (MAF)

INDELS

- INDEL = INsertion or DELetion
- Typically defined as changes of **1 to ~50 base pairs**
- Larger changes are often called **structural variants**
- Very small INDELs (1–3 bp) are common in coding regions



Structural Variants (SVs) are **larger-scale changes** in the genome structure — bigger than SNPs and INDELs

Deletion (DEL)	A large chunk of DNA is missing	Loss of tumor suppressor gene
Insertion (INS)	A large new segment is added	Viral DNA inserted into human genome
Duplication (DUP)	A segment is copied and repeated	Gene amplification (e.g., HER2 in breast cancer)
Inversion (INV)	A DNA segment is flipped in orientation	May disrupt genes at the breakpoints
Translocation (TRA)	A segment moves to a different chromosome	*BCR–ABL* fusion in CML (t[9;22])
Copy Number Variation (CNV)	Change in the number of gene copies	Can be gains or losses

Structural Variation

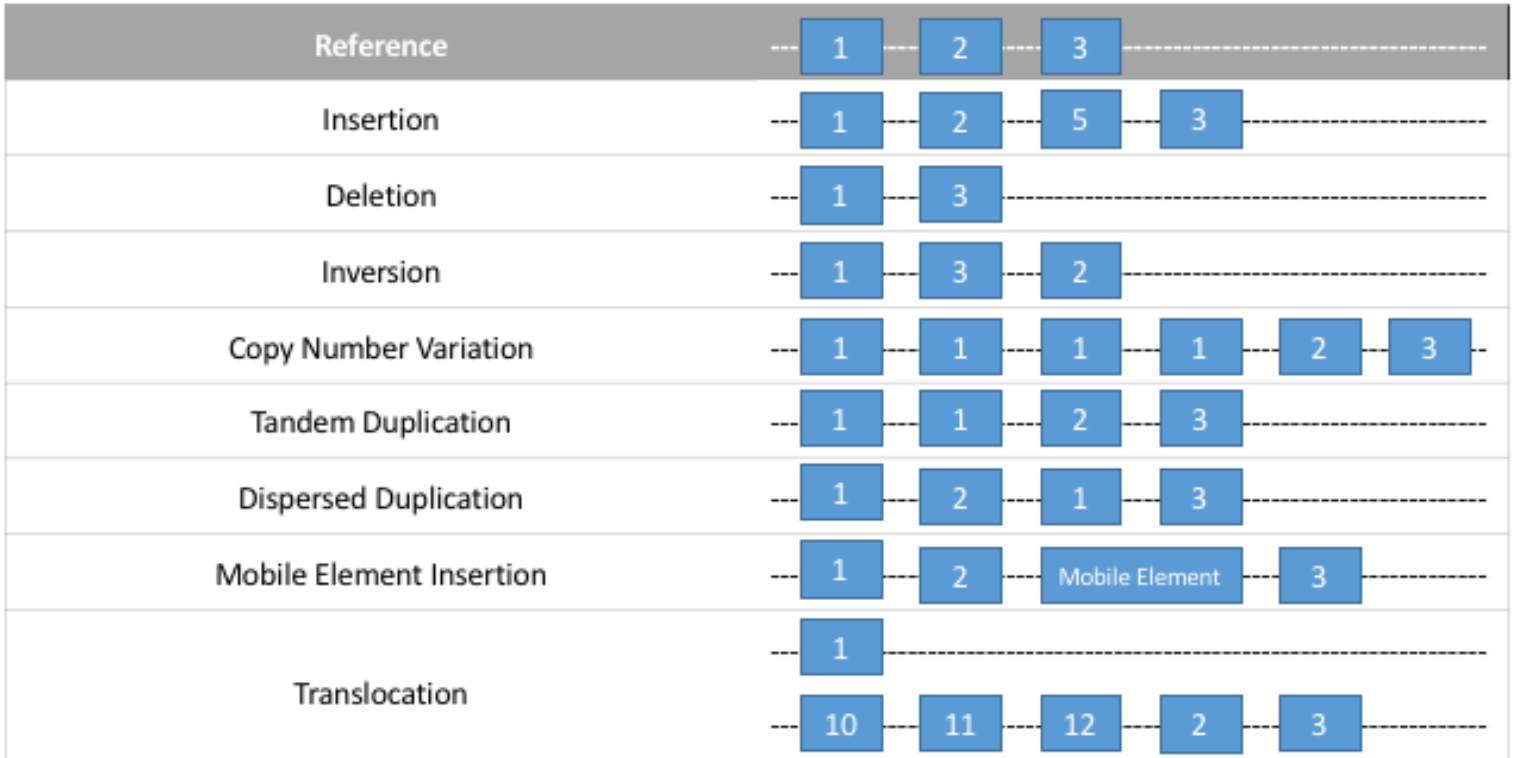
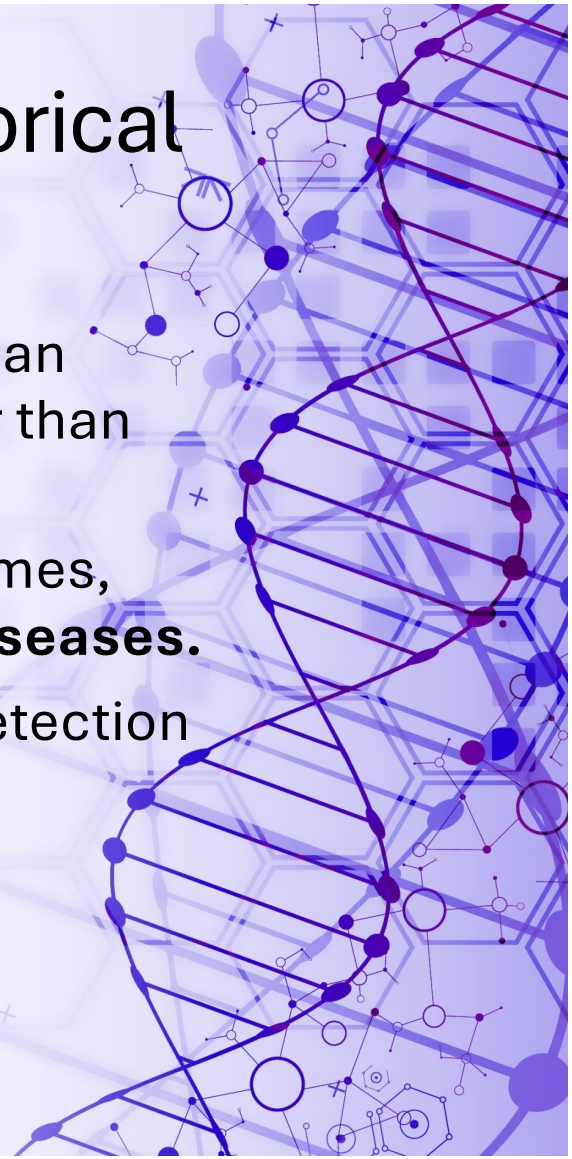


Figure 1: Depiction of different types of structural variants compared to the reference genome. Each different number represents a different gene.

From: https://upload.wikimedia.org/wikipedia/commons/2/27/Structural_Variation.png

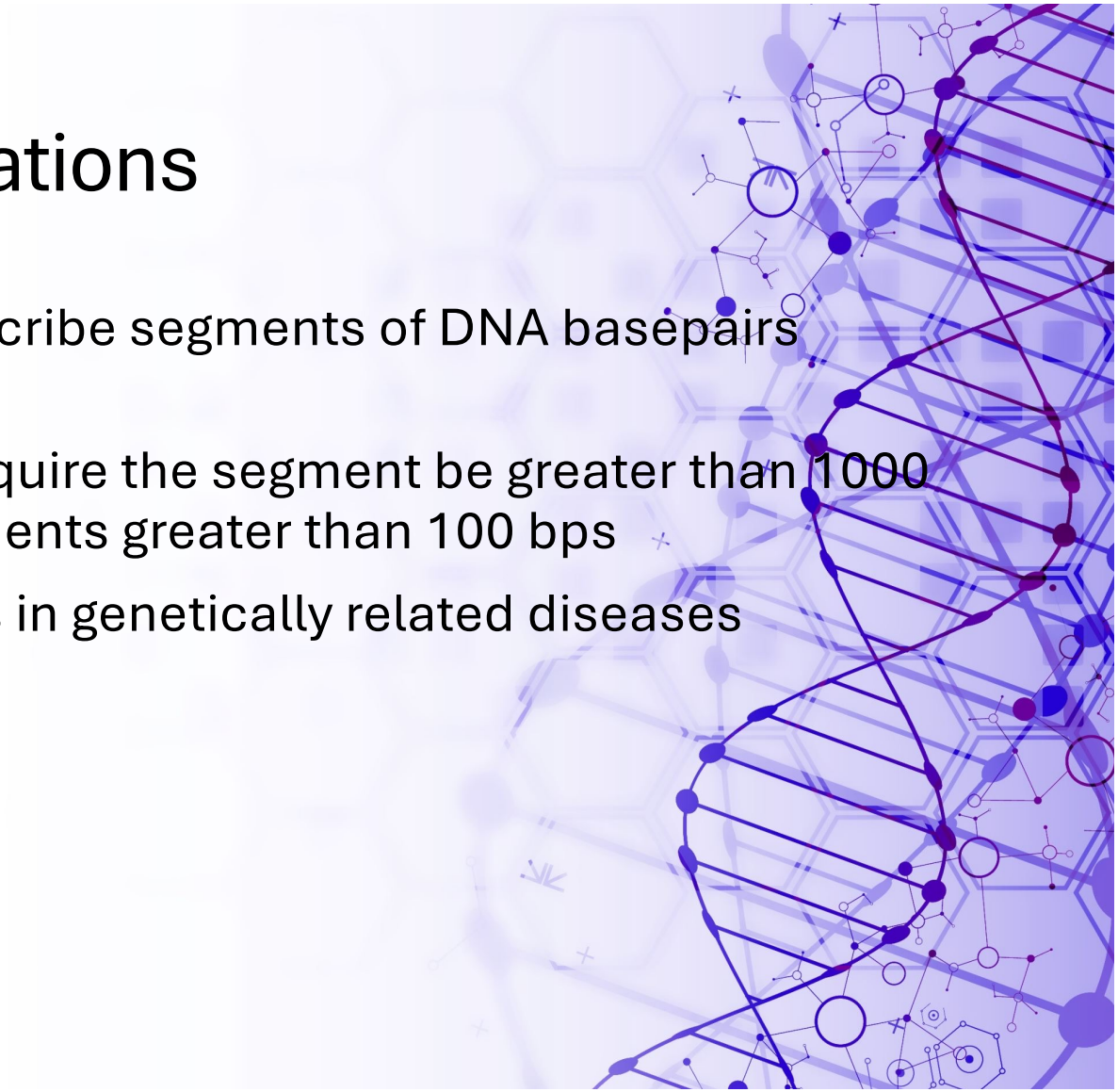
Genomic structural variation—historical context

- In earlier days (before Genome sequencing), one can observe large genomic structural variation (greater than 3Mbps) using a microscope.
- Observation of an abnormal number of chromosomes, called **Aneuploidy, is linked with many known diseases.**
- Large-scale Genomic sequencing allows for the detection of smaller structural variations.



Copy Number Variations

- A terminology used to describe segments of DNA basepairs amplified or deleted
- Earlier definition would require the segment be greater than 1000 bps, later defined as segments greater than 100 bps
- CNV plays important roles in genetically related diseases



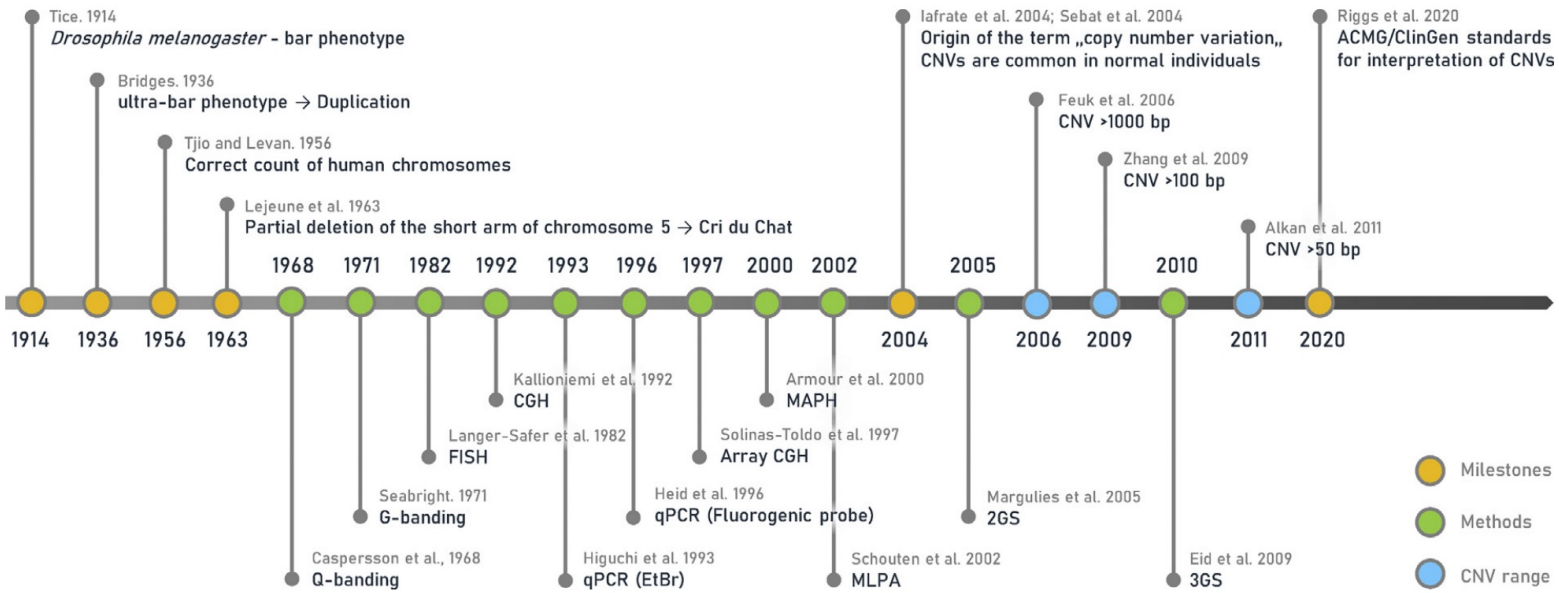


Fig. 2 Timeline of CNV research. It includes research milestones (orange) in the context of evolving methods for the assessment of CNVs (green) and the minimal length of variants to be considered as CNVs at a given time (blue). Main trends of CNV research are visible, e.g., the relatively wide time-frame following the first descriptions of large CNVs around the beginning of the 20th century. This nearly century-long phase was mainly about the development of methods, which finally allowed genome-wide, high-resolution CNV detection around the beginning of the 21st century, leading to the recognition of common features of CNVs and a subsequent shift in their definition, allowing shorter and shorter variants to be considered CNVs.

From review paper by Pos et al, 2021, DNA copy number variation: Main characteristics, evolutionary significance, and pathological aspects; https://doi.org/10.1016/j.bj.2021.02.003

What have we learned?-read a landmark paper

CANCERGENOMICS
www.sciencemag.org/special/cancergenomics

REVIEW

Cancer Genome Landscapes

Bert Vogelstein, Nickolas Papadopoulos, Victor E. Velculescu, Shibin Zhou, Luis A. Diaz Jr., Kenneth W. Kinzler*

Over the past decade, comprehensive sequencing efforts have revealed the genomic landscapes of common forms of human cancer. For most cancer types, this landscape consists of a small number of "mountains" (genes altered in a high percentage of tumors) and a much larger number of "hills" (genes altered infrequently). To date, these studies have revealed ~140 genes that, when altered by intragenic mutations, can promote or "drive" tumorigenesis. A typical tumor contains two to eight of these "driver gene" mutations; the remaining mutations are passengers that confer no selective growth advantage. Driver genes can be classified into 12 signaling pathways that regulate three core cellular processes: cell fate, cell survival, and genome maintenance. A better understanding of these pathways is one of the most pressing needs in basic cancer research. Even now, however, our knowledge of cancer genomes is sufficient to guide the development of more effective approaches for reducing cancer morbidity and mortality.

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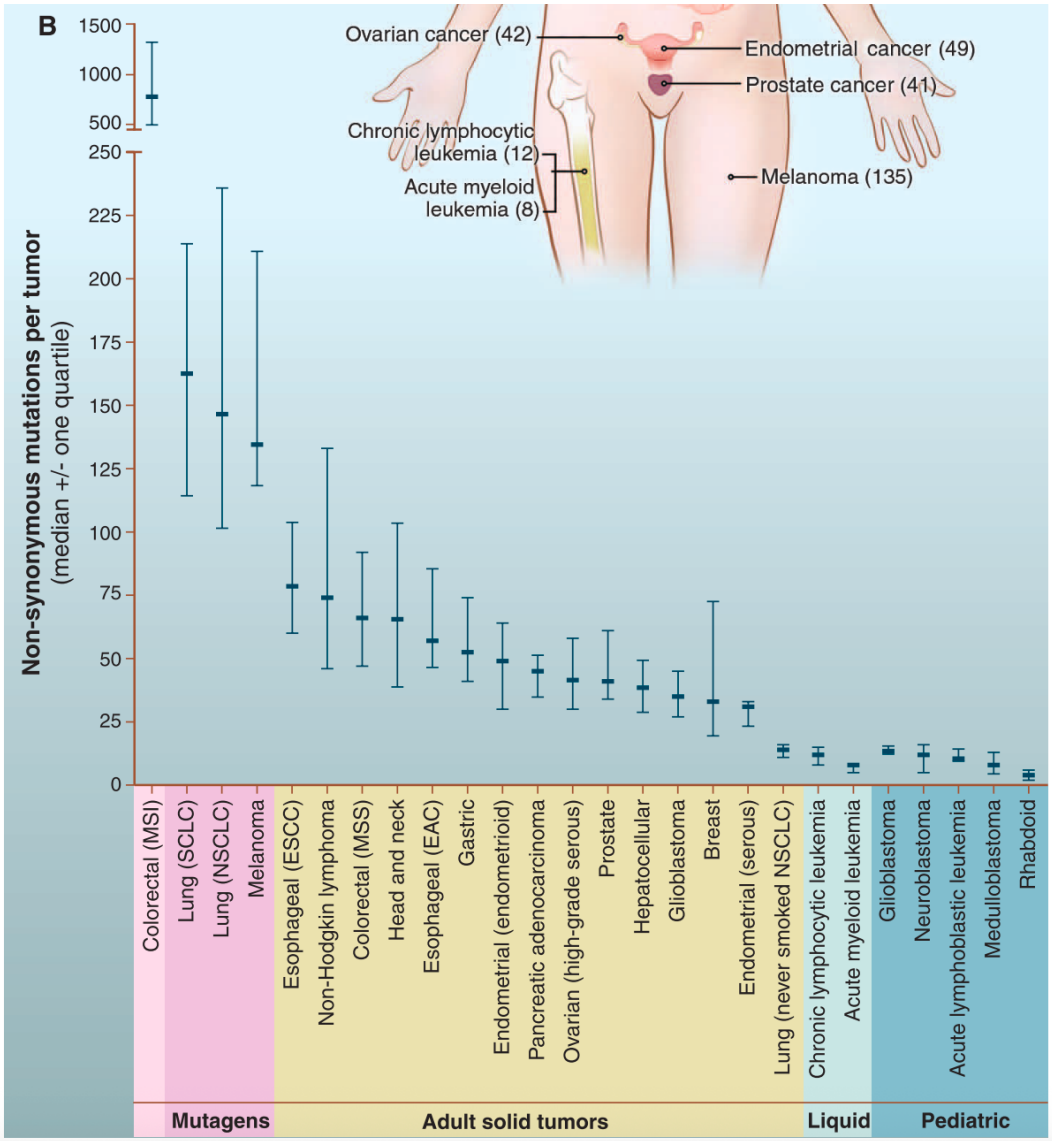
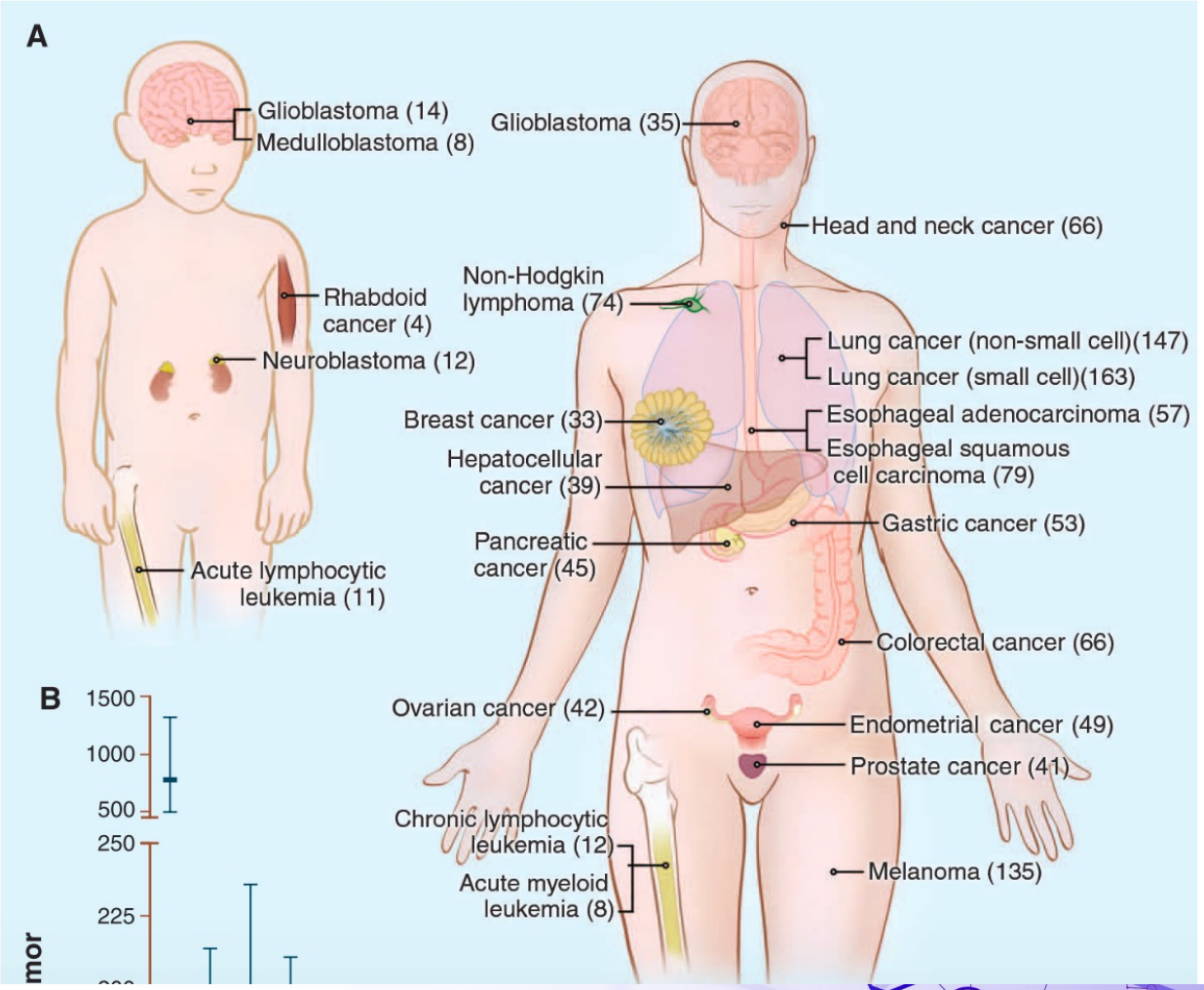
https://www.science.org/doi/10.1126/science.1235122

Number of mutations

- Number of mutations per tumor is shown on the right figure (Figure 1B from the paper)
- These mutations spread over 33 to 60 genes
- 95% of these mutations involve single base pair substitution (SBS) –which mostly lead to missense change (change of amino acids)



- **Numbers in parentheses indicate the median number of nonsynonymous mutations per tumor.**
- **Lung cancer and melanoma have higher number of nonsynonymous mutations –caused by known factors: smoke and UV-damage**



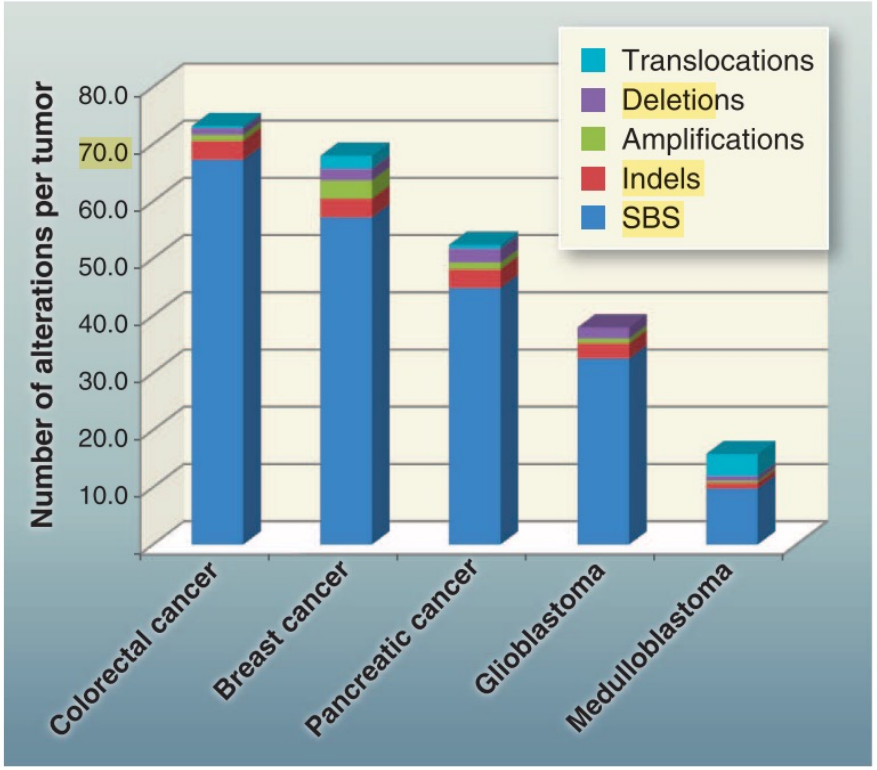


Fig. 3. Total alterations affecting protein-coding genes in selected tumors. Average number and types of genomic alterations per tumor, including single-base substitutions (SBS), small insertions and deletions (indels), amplifications, and homozygous deletions, as determined by genome-wide sequencing studies. For colorectal, breast, and pancreatic ductal cancer, and medulloblastomas, translocations are also included. The published data on which this figure is based are provided in table S1D.

- This shows the type of mutations that affect the tumors observed in tumors

Driver genes versus passenger genes?

- Driver genes are defined as mutations lead to selective growth benefits; also mutations occur more frequently than by chance
- Identifiable by “high number of mutations” in tumor samples
- Examples include: TP53, KRAS, PIK3CA
- Challenges exist in identifying driver versus passenger genes
- These authors used the concept:
  - Driver genes: A gene that contains driver gene mutations (Mut-Driver gene) or is expressed aberrantly in a fashion that confers a selective growth advantage (Epi-Driver gene).
  - Driver genes are further categorized to oncogenes versus tumor suppressor genes.

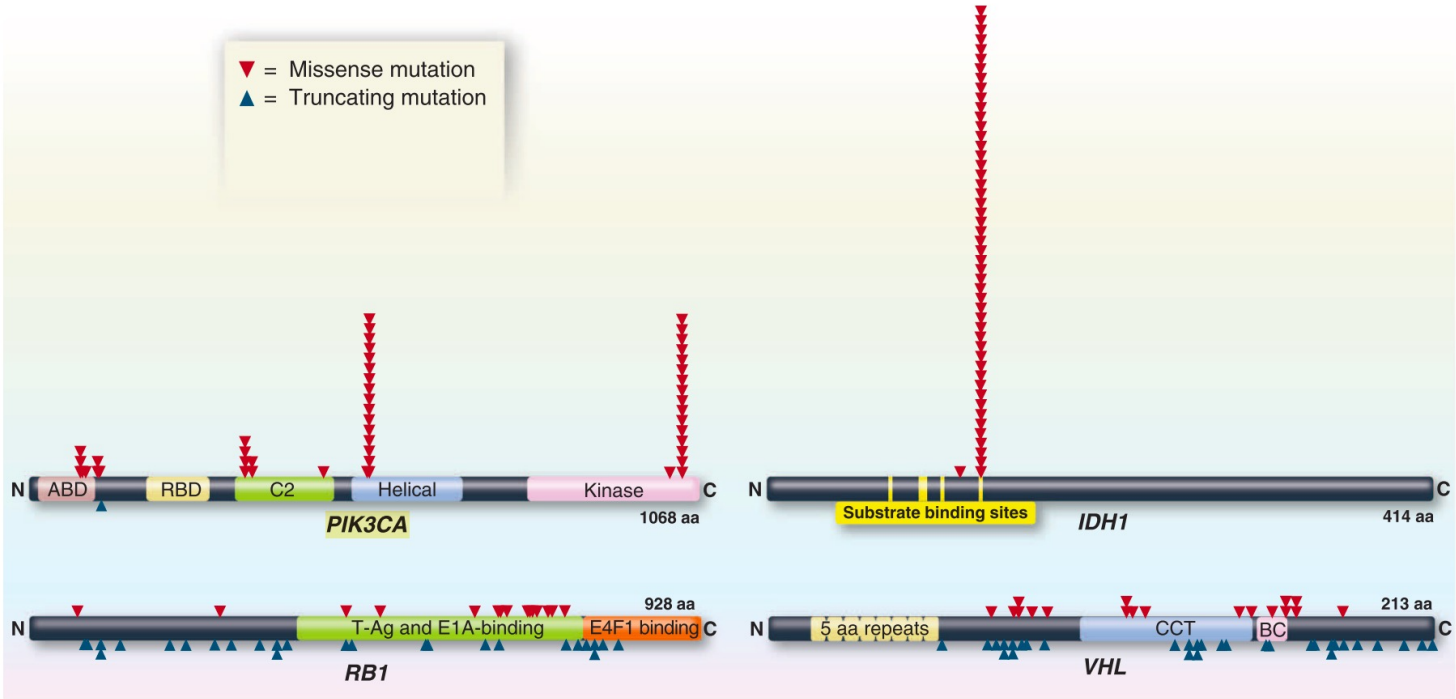


Fig. 4. Distribution of mutations in two oncogenes (*PIK3CA* and *IDH1*) and two tumor suppressor genes (*RB1* and *VHL*). The distribution of missense mutations (red arrowheads) and truncating mutations (blue arrowheads) in representative oncogenes and tumor suppressor genes are shown. The data were collected from genome-wide studies annotated in the COSMIC database (release version 61). For *PIK3CA* and *IDH1*, mutations obtained from the COSMIC database were randomized by the Excel RAND function, and the first 50 are shown. For *RB1* and *VHL*, all mutations recorded in COSMIC are plotted. aa, amino acids.

This figure illustrates different patterns of observed mutation frequencies on two types genes

(A) Oncogenes: PIK3CA, IDH1; (B) Tumor suppressor genes, RB1 and VHL

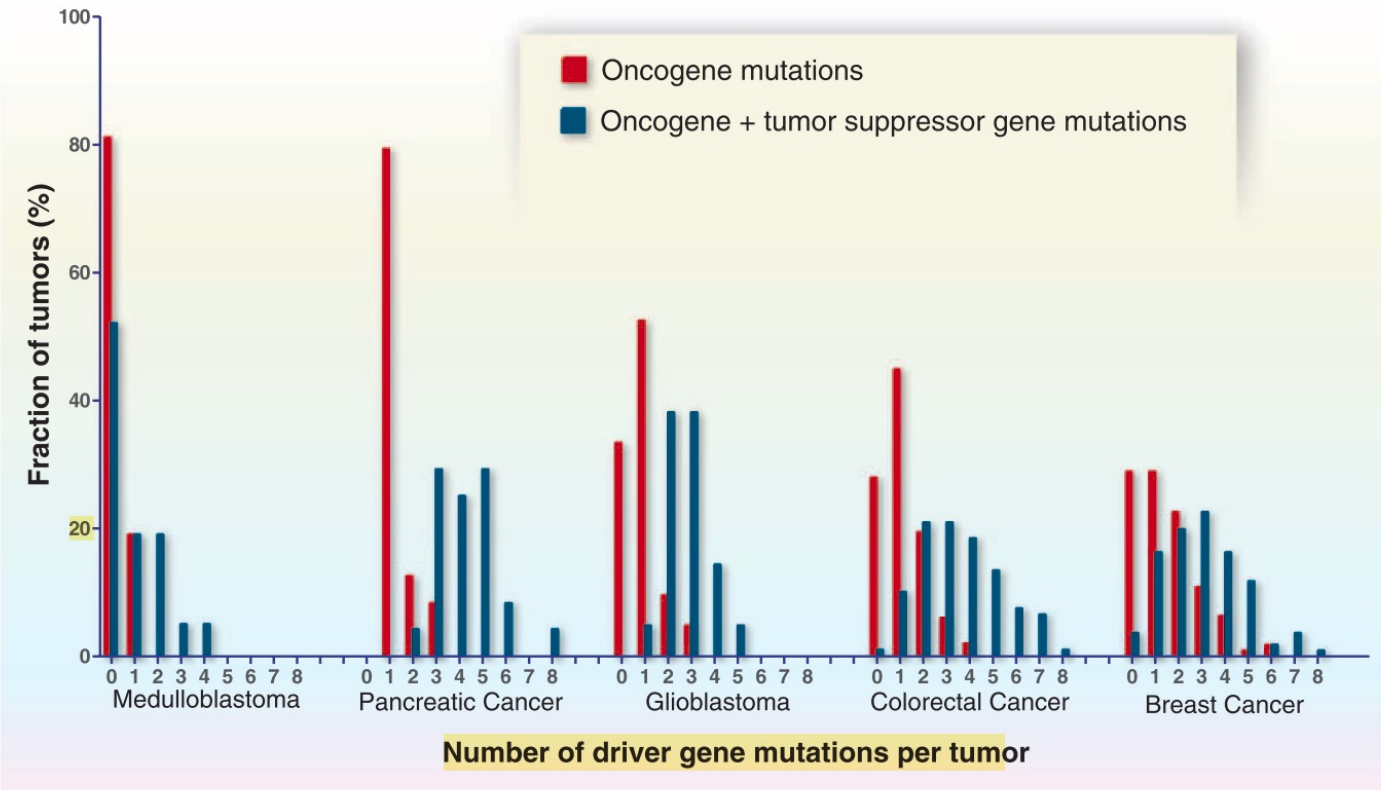


Fig. 5. Number and distribution of driver gene mutations in five tumor types. The total number of driver gene mutations [in oncogenes and tumor suppressor genes (TSGs)] is shown, as well as the number of oncogene mutations alone. The driver genes are listed in tables S2A and S2B. Translocations are not included in this figure, because few studies report translocations along with the other types of genetic alterations on a per-case basis. In the tumor types shown here, translocations affecting driver genes occur in less than 10% of samples. The published data on which this figure is based are provided in table S1E.

- Final summary

Box 2. Highlights

1. Most human cancers are caused by two to eight sequential alterations that develop over the course of 20 to 30 years.

2. Each of these alterations directly or indirectly increases the ratio of cell birth to cell death; that is, each alteration causes a selective growth advantage to the cell in which it resides.

3. The evidence to date suggests that there are ~140 genes whose intragenic mutations contribute to cancer (so-called Mut-driver genes). There are probably other genes (Epi-driver genes) that are altered by epigenetic mechanisms and cause a selective growth advantage, but the definitive identification of these genes has been challenging.

4. The known driver genes function through a dozen signaling pathways that regulate three core cellular processes: cell fate determination, cell survival, and genome maintenance.

5. Every individual tumor, even of the same histopathologic subtype as another tumor, is distinct with respect to its genetic alterations, but the pathways affected in different tumors are similar.

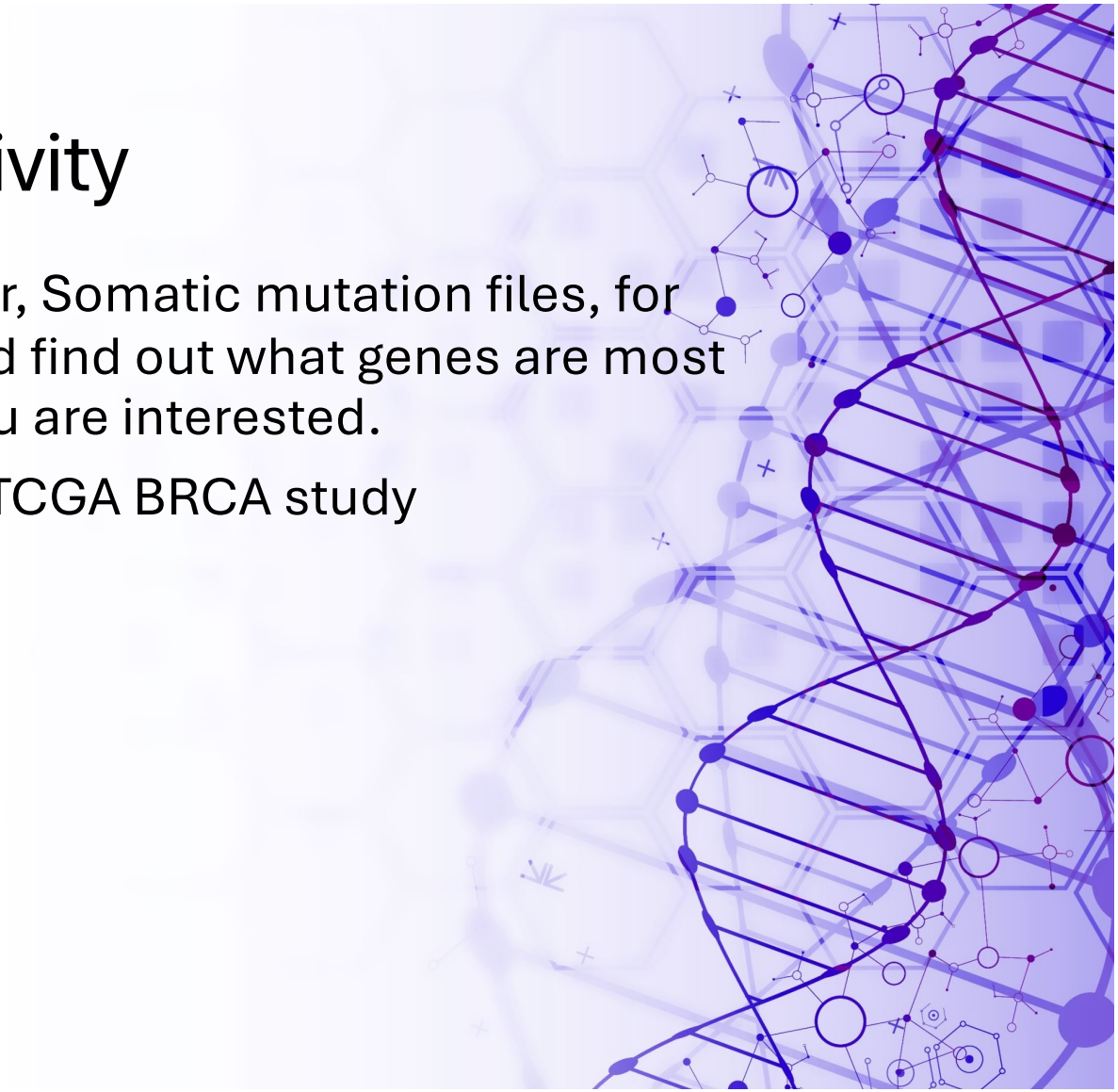
6. Genetic heterogeneity among the cells of an individual tumor always exists and can impact the response to therapeutics.

7. In the future, the most appropriate management plan for a patient with cancer will be informed by an assessment of the components of the patient's germline genome and the genome of his or her tumor.

8. The information from cancer genome studies can also be exploited to improve methods for prevention and early detection of cancer, which will be essential to reduce cancer morbidity and mortality.

Blogging Activity

- Write a blog that talks about what we have learned about mutations in tumor through genomic sequencing. Be sure to explain some technical terms such as SNPs, driver genes, onco genes. You can use 2013 Science paper by Volgestein et al as your primary reference sources
- Another landmark paper is “Lessons from the Cancer Genome”,in Cell, https://www.cell.com/cell/fulltext/S0092-8674(13)00288-2

Python Coding Activity

- Download at Xena Browser, Somatic mutation files, for one of the TCGA study, and find out what genes are most mutated for the cancer you are interested.
- In-class, we will focus on TCGA BRCA study